

The Indian story of VLSI began in 1985, when Texas Instruments set up the first design centre in Bangalore, thereby ushering in an era of wide-spread chip design and VLSI. Today, Texas Instruments finds itself in the company of names such as AMD, Intel and NVIDIA, apart from firms such as KPIT and Mindtree Consulting engaged in ASIC (Application Specific Integrated Circuit) designs. According to 6Wresearch, VLSI design service companies in India are expected to be worth \$3.02 billion by 2017, from all three segments — analogue, digital and mixed signal. However, the industry is expected to shift to mixed signal in the coming years. Also, despite the difference in salaries slowly narrowing, having designs produced in India still presents a huge comparative advantage; so the future is bright.

The Indian link to VLSI

Processors and chips developed today contain millions of transistors, hence, the term VLSI (very large scale integration). NVIDIA's Indian operation has more than 780 employees across three sites – Bangalore, Pune and Hyderabad. According to Amit Verma, Director of Human Resources, NVIDIA (India), “In these centres, the focus is on end-to-end ASIC product development encompassing architecture, RTL design, verification, timing, physical design, system engineering (including board bring up and debugging) and quality assurance for chipset and mobile products. NVIDIA's Indian engineers are part of the company's worldwide team and their contribution is vital to the business as a whole.”

On the other hand, AMD has two centres in India, where AMD teams usually work on one or two products at a time. This involves a fair bit of interaction with the teams abroad. “Shanghai was, for example, a joint effort between Boston and India”, said Karthik Muthusamy, former General Manager, India Design Centre at AMD India Engineering Centre.

Basic requirements

Passion and an early understanding of this field is essential. Aspirants will work with sophisticated design tools, chip architectures and test methods. So, apart from technical expertise, additional skill sets are also important. “Key areas that we look for in our engineers would be strong analytical skills, good problem solving skills, a willingness to work with teams as collaboration is a part of the job profile, and so a lot of communication skills.” said Rohit Vidwans, former director of Digital Enterprise Group (DEG), Intel India. Specific to chip design activities, companies look for